EstateEstimation

A Lab Project Report submitted for the partial fulfillment of the degree

of

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in

Computer Science and Engineering

Submitted by

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CERTIFICATE

This is to certify that the Lab Project entitled **"EstateEstimation"** submitted by **K.S D S S V S Varalakshmi (S220029)** and **Pratyusha Bisoi (S220479)** in partial fulfillment of the requirements for the Data Science with Python Lab Project.

Faculty Signature

ABSTRACT

House price prediction is an important application of data science that helps in understanding the factors influencing real estate values. This project focuses on developing machine learning models to **predict** house prices using a structured housing dataset. The dataset was **preprocessed** by **handling missing values**, performing **data cleaning**, converting **categorical features into numerical format**, and applying **feature engineering techniques** such as deriving house age and price per square foot.

Various data visualization techniques were used to analyze feature distributions and relationships, providing insights into key factors affecting house prices, including living area, number of rooms, and location. Multiple **regression models**, including **Linear Regression, Polynomial Regression, Random Forest, and Gradient Boosting**, were implemented and evaluated using performance **metrics** such as Mean Squared Error **(MSE)**, Root Mean Squared Error **(RMSE)**, and **R^2 score**.

The results show that **Linear Regression** achieved an **R^2 score** of **0.90** with an **RMSE** of **63,406**, while **Polynomial Regression** achieved an **R^2 score of 0.999** with an **RMSE** of **265.9**, indicating possible **overfitting**. The **Random Forest Regressor** provided the **best performance** with a test **R^2 score** of **0.979** and RMSE of **21,669**, followed by **Gradient Boosting** with an **R^2 score** of **0.974** and **RMSE of 26,273**.

Overall, the project demonstrates the importance of data preprocessing, visualization, and model selection in building accurate predictive systems. The results highlight that ensemble methods are more effective for this task and can be successfully applied to real-world house price prediction problems.

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Chapter 1

Introduction

This chapter presents an overview of the project including background,problem statement,objectives and organization of the report.

1.1 Background

In recent years, machine learning has become an important tool for solving real-world problems. One such application is house price prediction, where various factors influence the final price. This project applies data science techniques such as preprocessing, visualization, and model building to analyze housing data and predict prices effectively.

House price prediction is an important application of data science. Prices depend on multiple factors such as living area, number of bedrooms, location, and facilities. Machine learning helps to analyze these features and predict house prices effectively.

1.2 Problem Statement

The objective of this project is to predict house prices based on various features available in the dataset using machine learning techniques.

1.3 Objectives

The Objectives of this project are as follows:

- To clean and preprocess the housing dataset
- To perform data visualization for better understanding
- To build a machine learning model for prediction
- To evaluate model performance

1.4 Organisation of Report

The report is divided into data preprocessing, visualization, model fitting, and evaluation of results.

Chapter 2

Data Preprocessing

Data preprocessing is a critical step in the machine learning pipeline, as the quality of input data directly impacts the performance of the model. In this project, extensive preprocessing techniques were applied to clean, transform, and prepare the housing dataset for analysis and model building.

2.1 Data Exploration

Initially, the dataset was explored to understand its structure, feature types, and statistical properties. The following methods were used:

- `head()` to preview sample records

```
df.head() #first 5 rows
```

Python

	price	bedrooms	bathrooms	livingArea	lotSize	yearBuilt	propertyTaxRate	taxAssessedValue	zestimate	rentZestimate	latitude	longitude	resoFacts/stories
0	315400	3.0	3.0	1806.0	1555.0	2007.0	1.12	273960.0	317406.0	1950.0	45.543571	-122.441765	NaN
1	400000	3.0	3.0	1518.0	3484.0	2001.0	1.12	NaN	NaN	NaN	45.547585	-122.453224	2.0
2	512000	4.0	3.0	2724.0	9583.0	1982.0	1.12	397560.0	512355.0	1995.0	45.488228	-122.444420	2.0
3	348000	3.0	1.0	1217.0	13939.0	1953.0	1.12	269520.0	NaN	1695.0	45.526634	-122.464088	1.0
4	510000	6.0	3.0	2400.0	8545.0	1967.0	1.12	421980.0	NaN	1395.0	45.511238	-122.431534	NaN

Figure 2.1: Sample records of the dataset

- `info()` to examine data types and non-null counts

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 25731 entries, 0 to 25730
Data columns (total 19 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   price                       25731 non-null  int64
 1   bedrooms                    24961 non-null  float64
 2   bathrooms                   25247 non-null  float64
 3   livingArea                  25266 non-null  float64
 4   lotSize                     22841 non-null  float64
 5   yearBuilt                   25185 non-null  float64
 6   propertyTaxRate             25720 non-null  float64
 7   taxAssessedValue            24337 non-null  float64
 8   zestimate                   23757 non-null  float64
 9   rentZestimate               24826 non-null  float64
 10  latitude                    25718 non-null  float64
 11  longitude                   25718 non-null  float64
 12  resoFacts/stories           21206 non-null  float64
 13  resoFacts/hasGarage         25731 non-null  bool
 14  resoFacts/hasHeating        25730 non-null  object
 15  resoFacts/hasCooling        21235 non-null  object
 16  resoFacts/hasFireplace      21107 non-null  object
 17  resoFacts/isNewConstruction 21374 non-null  object
 18  homeType                    25731 non-null  str
dtypes: bool(1), float64(12), int64(1), object(4), str(1)
```

Figure 2.2: Dataset Structure and Data types using `info()` function

- `describe()` to analyze statistical summaries

This step helped identify:

- Numerical and categorical features
- Missing values
- Potential anomalies and outliers

2.2 Handling Missing Values

Missing values were detected using:

- `isnull()` and `sum()` functions

```
df.isnull().sum() #col wise null count
```

```
price                          0
bedrooms                     770
bathrooms                    484
livingArea                   465
lotSize                     2890
yearBuilt                    546
propertyTaxRate               11
taxAssessedValue            1394
zestimate                   1974
rentZestimate                905
latitude                      13
longitude                     13
resoFacts/stories           4525
resoFacts/hasGarage            0
resoFacts/hasHeating           1
resoFacts/hasCooling        4496
resoFacts/hasFireplace      4624
resoFacts/isNewConstruction 4357
homeType                       0
dtype: int64
```

Figure 2.3: Missing values present in the dataset before preprocessing

The following strategies were applied:

- Numerical columns were filled using median values to reduce the impact of skewness and outliers
- Any inconsistent or undefined values were handled appropriately

```
df.isnull().sum() #nulls filled
```

```
price                        0
bedrooms                     0
bathrooms                    0
livingArea                   0
lotSize                      0
yearBuilt                    0
propertyTaxRate              0
taxAssessedValue             0
zestimate                    0
rentZestimate                0
latitude                     0
longitude                    0
resoFacts/stories            0
resoFacts/hasGarage          0
resoFacts/hasHeating         0
resoFacts/hasCooling         0
resoFacts/hasFireplace       0
resoFacts/isNewConstruction  0
homeType                     0
dtype: int64
```

Figure 2.4: Dataset after handling missing values

This ensured that no null values remained in the dataset before model training.

2.3 Data Cleaning and Consistency Checks

To improve data quality:

- Duplicate entries (if any) were checked and removed
- Irregular or inconsistent values were corrected

- Feature names and formats were standardized

Additionally:

- Infinite values (`inf`, `-inf`) were identified and replaced with appropriate values
- Dataset integrity was verified after cleaning

2.4 Data Type Conversion

Machine learning models require numerical input. Therefore:

- Categorical and boolean features such as heating, cooling, fireplace, and new construction were converted into binary numerical format (0 and 1)

```
df.dtypes

price                          int64
bedrooms                     float64
bathrooms                    float64
livingArea                   float64
lotSize                      float64
yearBuilt                    float64
propertyTaxRate              float64
taxAssessedValue             float64
zestimate                    float64
rentZestimate                float64
latitude                     float64
longitude                    float64
resoFacts/stories            float64
resoFacts/hasGarage            int64
resoFacts/hasHeating           int64
resoFacts/hasCooling           int64
resoFacts/hasFireplace         int64
resoFacts/isNewConstruction    int64
homeType                         str
dtype: object
```

Figure 2.5: Dataset after converting categorical features into numerical format

This transformation ensured compatibility with regression models.

2.5 Feature Engineering

To enhance model performance and extract meaningful insights, new features were created:

- **House Age:** Derived from construction year. Helps capture depreciation and property condition

- **Price per Square Foot:** Calculated as:

$$\text{Price per Square Foot} = \frac{\text{Price}}{\text{Living Area}}$$

  Provides normalized price comparison

homeType_CONDO	homeType_HOME_TYPE_UNKNOWN	homeType_SINGLE_FAMILY	homeType_TOWNHOUSE	house_age	price_per_sqft
0	0	0	1	18.0	174.640089
0	0	1	0	24.0	263.504611
0	0	1	0	43.0	187.958884
0	0	1	0	72.0	285.949055
0	0	0	0	58.0	212.500000

Figure 2.6: Newly engineered features: House Age and Price per Square Foot

These engineered features improved the predictive capability of the model by capturing hidden relationships in the data.

2.6 Outlier Detection and Removal

Outliers can significantly affect model accuracy. Therefore:

- Extreme values in the target variable (price) were identified
- The top 5% highest price values were removed using quantile-based filtering

This helped:

- Reduce skewness
- Improve model generalization
- Prevent bias toward extreme values

2.7 Feature Scaling

To ensure all features contribute equally to the model:

- Numerical features were scaled using normalization techniques

2.8 Data Validation

After preprocessing, the dataset was validated to ensure:

- No missing values remained
- All features were in correct format
- No infinite or inconsistent values existed

2.9 Summary of Preprocessing Steps

The preprocessing pipeline included:

- Data exploration
- Handling missing values
- Data cleaning and validation
- Data type conversion
- Feature engineering
- Outlier removal
- Feature scaling

These steps ensured high-quality input data, leading to better model performance and reliable predictions.

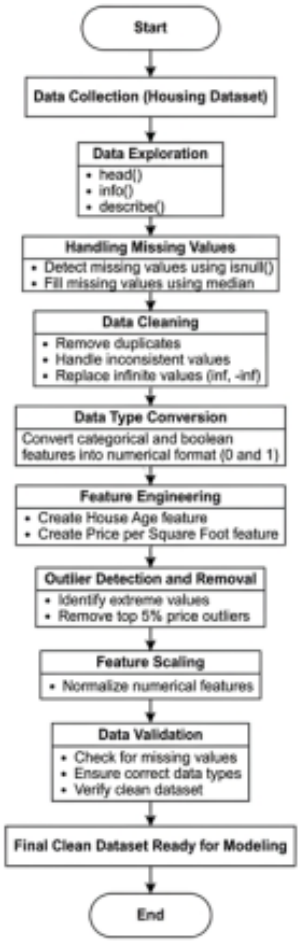

Figure 2.7: Flow Chart of Data Preprocessing

Chapter 3

Data Visualization

Data visualization is a crucial step in data analysis that helps in understanding the distribution, patterns, relationships, and anomalies present in the dataset. It provides meaningful insights that guide feature selection and model building.

In this project, various visualization techniques were used to analyze the housing dataset and understand the factors influencing house prices.

3.1 Univariate Analysis

Univariate analysis focuses on analyzing individual features independently to understand their distribution and characteristics.

The following techniques were used:

Histograms were used to analyze the distribution of numerical features such as house prices. The price distribution showed a right-skewed pattern, indicating that most houses fall within a lower price range, with a few high-priced outliers.

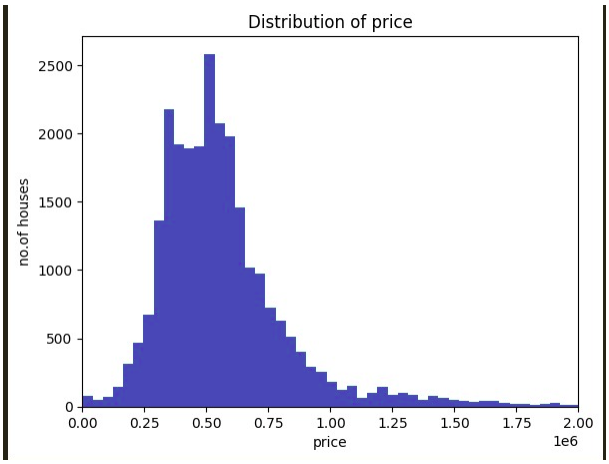

Figure 3.1: Distribution of house prices using histogram

Box Plots were used to detect outliers in features like price. These plots clearly highlighted the presence of extreme values, which were later handled during preprocessing.

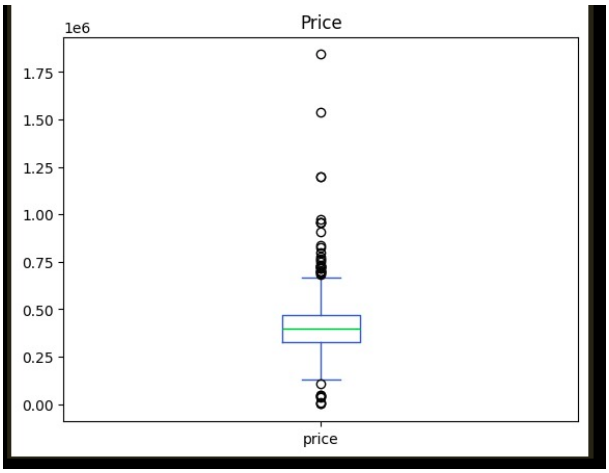

Figure 3.2: Box plot showing outliers in house prices

Bar Charts were used to visualize categorical features, helping in understanding the frequency distribution of different categories.

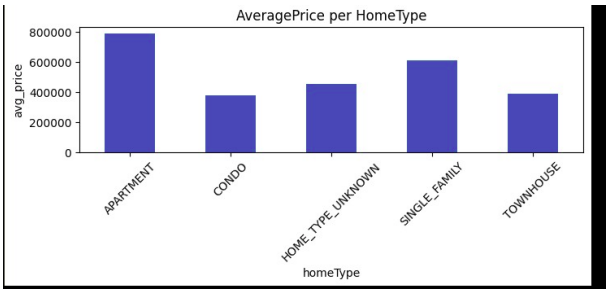

Figure 3.3: Bar chart representing categorical feature distribution

This analysis helped in identifying skewness, spread, and outliers in the dataset.

3.2 Bivariate Analysis

Bivariate analysis examines the relationship between two variables, especially between input features and the target variable (price).

The following visualizations were used:

Scatter Plots were used to analyze the relationship between living area and house price. A strong positive correlation was observed, indicating that houses with larger living areas tend to have higher prices.

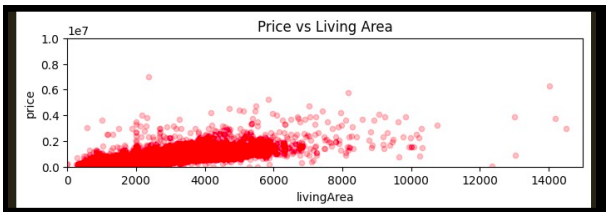

Figure 3.4: Scatter plot showing relationship between living area and house price

Line Plots were used to observe trends in price values across the dataset. Although not time-based, they helped in identifying fluctuations and irregular patterns.

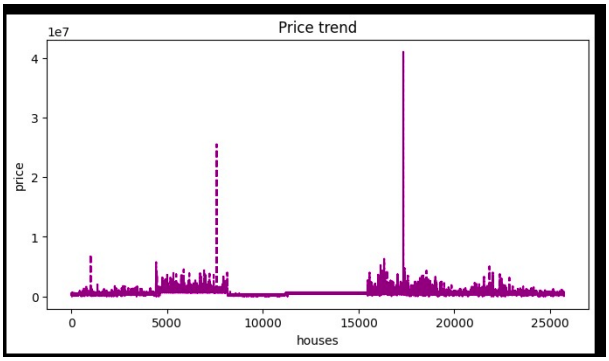

Figure 3.5: Line plot showing variation of house prices

These plots helped in understanding which features have a significant impact on house prices.

3.3 Feature Relationship Analysis

To further understand interactions between multiple features:

Relationships between features such as number of bedrooms, bathrooms, and price were analyzed.

Figure 3.6: Relationship between number of bedrooms and house price

These visualizations showed that an increase in the number of bedrooms and bathrooms generally leads to an increase in house price, although with some variability.

This analysis helped in identifying important features for model training.

3.4 Advanced Visualization

Advanced visualization techniques were used to extract deeper insights:

Geographical Visualization (Choropleth Map) was used to represent variation in house prices across different regions. This helped in understanding location-based price differences.

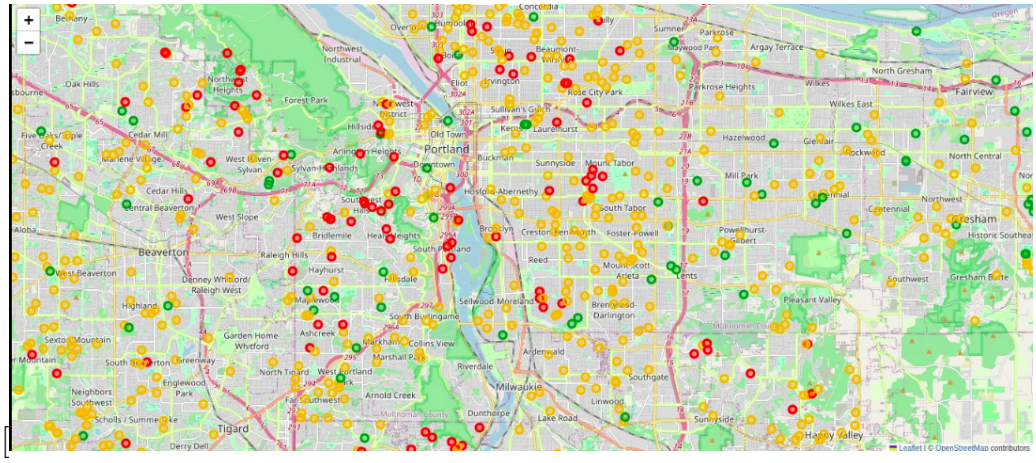

Figure 3.7: Choropleth map illustrating the geographical variation in the house prices across different regions,highlighting location based price differences.Image is linked to the interactive visualization.Click on the image for detailed analysis.

Regions with higher demand showed higher property prices, indicating that location is a significant factor in price prediction.

3.5 Insights from Visualization

The following key insights were derived from the visual analysis:

- House prices are mostly concentrated in lower ranges with a few high-value outliers
- Living area has a strong positive correlation with price
- Larger houses tend to have higher prices
- Location plays an important role in determining house prices
- Newer houses generally have higher prices compared to older ones

These insights helped in selecting relevant features and improving model performance.

3.6 Limitations of Visualization

Although visualization provides valuable insights, it has certain limitations:

- Some relationships may not be clearly visible due to data complexity
- Geographical visualization depends on the availability of accurate location data
- Visual interpretations may sometimes be subjective

Despite these limitations, visualization played a key role in understanding the dataset and guiding further analysis.

3.7 Summary of Visualization Techniques

The visualization process included:

- Univariate analysis (histograms, box plots, bar charts)
- Bivariate analysis (scatter plots, line plots)
- Feature relationship analysis
- Advanced visualization (choropleth maps)

These techniques provided a comprehensive understanding of the dataset and supported effective model development.

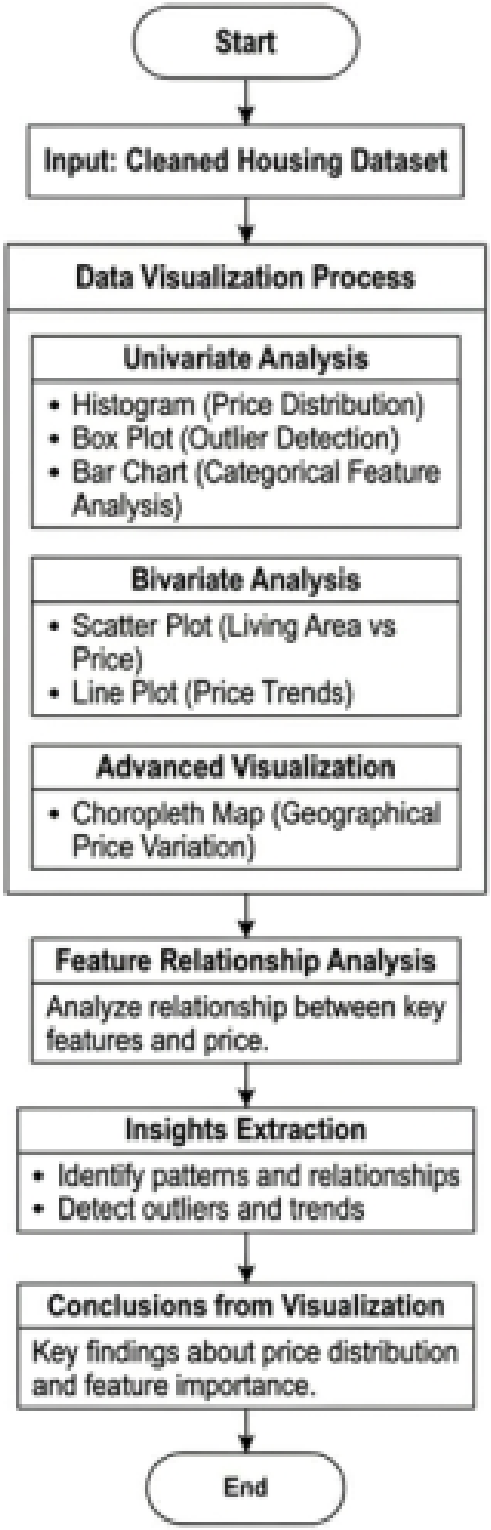

Figure 3.8: Flow Chart of Data Visualization Process

Chapter 4

Model Fitting

4.1 Introduction to Model Fitting

Model fitting is a crucial step in the machine learning pipeline, where algorithms are trained on historical data to learn patterns and relationships between input features and the target variable. In this project, multiple regression models were implemented to predict house prices based on various property attributes. The objective is to compare different models and identify the one that provides the most accurate and reliable predictions.

4.2 Data Preparation and Splitting

The dataset used for model training was preprocessed and cleaned prior to model fitting. The following preprocessing steps were applied:

- The dataset was loaded from a cleaned CSV file
- Categorical features such as home type were converted into numerical form using one-hot encoding
- Boolean features were transformed into integer values
- Missing values were handled using median imputation
- Infinite values were replaced with appropriate values
- New features such as house age and price per square foot were created
- The top 5% of extreme price values were removed to eliminate outliers

After preprocessing, the dataset was divided into:

- Input features (X): All independent variables
- Target variable (y): House price

The dataset was then split into:

- 80% training data

- 20% testing data

This ensures that the model is evaluated on unseen data, which is essential for measuring its generalization capability.

4.3 Models Implemented

The following regression models were used in this project:

4.3.1 Linear Regression

Linear Regression is a fundamental supervised learning algorithm that models the relationship between input features and the target variable using a linear equation. It serves as a baseline model for comparison.

4.3.2 Polynomial Regression (Degree 2)

Polynomial Regression extends the linear model by introducing higher-degree terms and interactions between features. This allows the model to capture nonlinear relationships in the data.

4.3.3 Random Forest Regressor

Random Forest is an ensemble learning technique that constructs multiple decision trees and aggregates their predictions. It is effective in capturing complex patterns and reducing overfitting.

4.3.4 Gradient Boosting Regressor

Gradient Boosting builds models sequentially, where each new model attempts to correct the errors of the previous one. It is known for its strong predictive performance on structured datasets.

4.4 Model Training Process

Each model was trained using the training dataset:

- Linear and Polynomial Regression models were trained using least squares optimization
- Polynomial features were generated for the polynomial model
- Random Forest and Gradient Boosting models were trained using ensemble learning techniques

Predictions were generated for both training and testing datasets to evaluate model performance.

4.5 Evaluation Metrics

The performance of the models was evaluated using the following metrics:

- Mean Squared Error (MSE): Measures the average squared difference between actual and predicted values
- Root Mean Squared Error (RMSE): Represents the prediction error in the same units as house price
- R^2 Score: Indicates how well the model explains the variance in the target variable

These metrics help in assessing both accuracy and reliability of the models.

4.6 Model Performance

The performance of each model is summarized below:

Linear Regression

- Train R^2 0.90
- Test R^2 0.90
- RMSE 63,406
- Provides a reasonable baseline but exhibits relatively higher prediction error

Polynomial Regression (Degree 2)

- Train R^2 0.999
- Test R^2 0.999
- RMSE 265.9
- Extremely high accuracy, but likely overfitting due to excessive model complexity

Random Forest Regressor

- Train R^2 0.996
- Test R^2 0.979
- RMSE 21,669
- Demonstrates the best generalization and lowest prediction error

Gradient Boosting Regressor

- Train R^2 0.982
- Test R^2 0.974
- RMSE 26,273
- Strong performance, slightly below Random Forest

4.7 Summary of Model Results

The performance metrics of all models are summarized in Table 4.1.

Table 4.1: Comparison of Model Performance

Model	**Train R^2**	**Test R^2**	**RMSE**
Linear Regression	0.90	0.90	63,406
Polynomial Regression	0.999	0.999	265.9
Random Forest	0.996	0.979	21,669
Gradient Boosting	0.982	0.974	26,273

The table clearly shows that ensemble models outperform traditional regression methods.

4.8 Model Comparison

Among all the models:

- Random Forest achieved the best performance with the lowest error and strong generalization
- Gradient Boosting also performed well but slightly below Random Forest
- Linear Regression served as a baseline model
- Polynomial Regression showed unrealistically high performance, indicating potential overfitting

Thus, Random Forest is identified as the most suitable model for this task.

4.9 Key Observations

The following observations were made:

- Ensemble models significantly outperform simple regression models
- Feature engineering improves model accuracy

- Outlier removal enhances generalization
- Polynomial models may lead to overfitting despite high accuracy
- Random Forest provides the best balance between bias and variance

4.10 Conclusion of Model Fitting

The model fitting process successfully evaluated multiple regression techniques for house price prediction.

Among all models:

- Random Forest Regressor emerged as the best-performing model
- It achieved high accuracy and strong generalization on unseen data

This project highlights the importance of:

- Proper data preprocessing
- Model comparison
- Avoiding overfitting

Future improvements may include:

- Hyperparameter tuning
- Cross-validation
- Exploring advanced ensemble technique

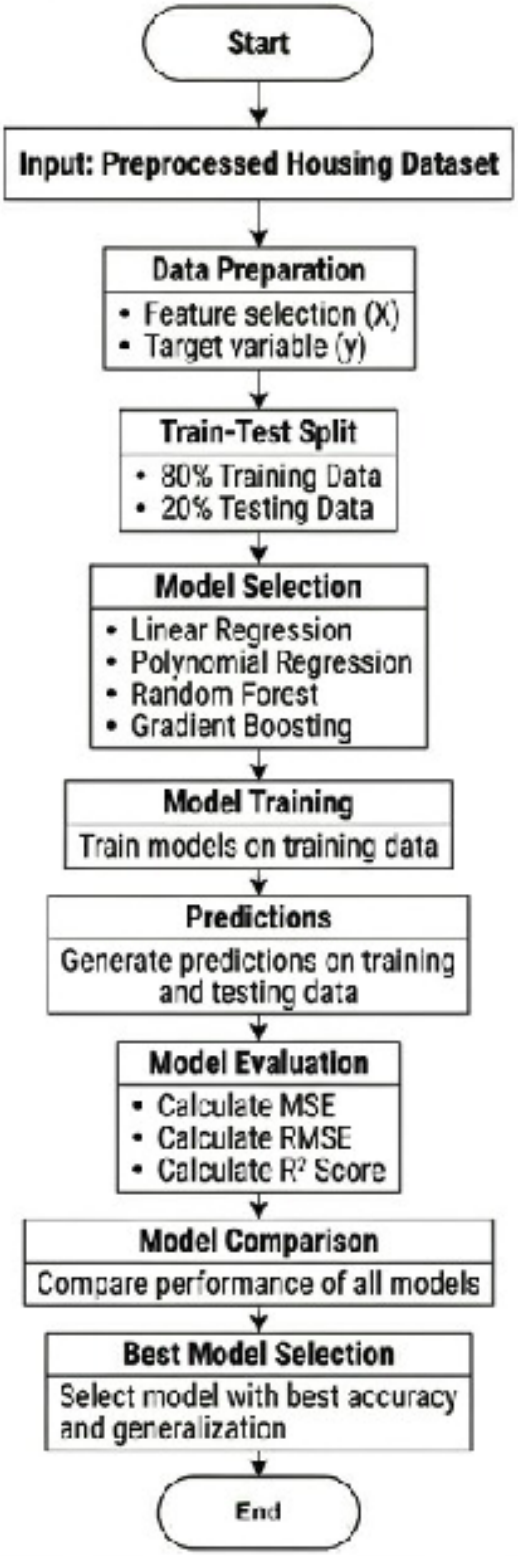

Figure 4.1: Flow Chart of Model Fitting

Final Conclusion

This project successfully demonstrated the application of data science and machine learning techniques for house price prediction. The dataset was thoroughly preprocessed through handling missing values, feature engineering, outlier removal, and scaling to ensure high-quality input for modeling.

Data visualization provided meaningful insights into feature distributions and relationships, highlighting key factors such as living area, location, and number of rooms that significantly influence house prices. Multiple regression models were implemented and evaluated, including Linear Regression, Polynomial Regression, Random Forest, and Gradient Boosting.

Among these, the Random Forest Regressor delivered the best performance, achieving a strong balance between accuracy and generalization. The results emphasize the importance of proper preprocessing, feature selection, and model comparison in building reliable predictive systems.

Overall, this project provides a practical understanding of the end-to-end machine learning workflow and demonstrates how data-driven approaches can be effectively used for real-world prediction problems. Future work can focus on hyperparameter tuning, cross-validation, and exploring more advanced models to further improve performance.

Chapter 5

References

Click On The Below Links To Access The project Files.

- Housing Dataset(kaggle)
- Cleaned Housing Dataset
- Data Preprocessing
- Data Visualization
- Model Fitting